

Janak Kabra

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EDUCATION

Vellore Institute of Technology

Bachelor of Technology in Computer Science and Engineering

Vellore, Tamil Nadu

Aug. 2024 – May 2028

EXPERIENCE

Web Developer (Freelance)

Jun. 2025 – Present

Self-Employed

Remote

- Designed and shipped responsive, mobile-first web apps for clients using React, Vite, and Node.js, owning the full cycle from build to deployment.
- Built a reusable Tailwind + Framer Motion component library, cutting duplicated markup across client sites and shortening new-page development time.
- Improved front-end performance through lazy-loading and code-splitting, reaching 95+ Google Lighthouse performance scores.

AI Research Intern

Oct. 2025 – Dec. 2025

Mythus AI Studios

Remote

- Cleaned and validated heterogeneous unstructured datasets feeding image-classification models, raising data quality ahead of production training.
- Rewrote feature-selection scripts with vectorized NumPy and multiprocessing, cutting pipeline runtime by roughly 20%.
- Reproduced computer-vision and neural-network architectures from published papers in PyTorch, building an internal reference notebook the team reused.

AI Domain Intern

Feb. 2026 – Jun. 2026

Infosys Springboard

Remote

- Built and benchmarked LSTM and multivariate-regression models for an agricultural time-series forecasting task, improving out-of-sample RMSE over the regression baseline.

PROJECTS

Volatility Surface Engine | GitHub | Live Demo | Python, Numba, FastAPI, Docker

May 2026

- Built an options pricing and risk engine from first principles, implementing a Heston stochastic-volatility Fourier solver and an arbitrage-free SVI implied-volatility surface model.
- Accelerated integration and optimizer loops with Numba JIT to reach sub-millisecond pricing, and resolved numerical instability (the “Little Heston Trap”) at long maturities using branch-cut-free representations of the complex logarithm.
- Containerized a FastAPI service with Docker to stream 3D Plotly volatility surfaces and option Greeks to a dashboard.

Deterministic Algo Debugger | GitHub | Live Demo | React, Zustand, HTML5 Canvas

Apr. 2026

- Built a web-based algorithm visualizer and step debugger that executes sorting and pathfinding algorithms in a virtual container, generating immutable execution snapshots for Big-O analysis.
- Engineered a custom HTML5 Canvas renderer holding a stable 60 FPS under workloads of 1,000+ elements.
- Added a dual-algorithm race mode to compare execution speed under identical inputs.

NSE Stock Screener | GitHub | Live Demo | FastAPI, PostgreSQL, React

Feb. 2026

- Built a fundamental stock screener for NIFTY 500 equities returning multi-factor query results in under 500ms.
- Designed database indexing strategies and compiled chained filters into single-roundtrip SQL queries to minimize latency.
- Built interactive charts and sparklines (Recharts) for clean visual analysis of quarterly ROE and P/E trends.

TECHNICAL SKILLS

Languages (Core): Python, TypeScript/JavaScript (ES6+), SQL

Languages (Familiar): C++, C, Java, Go, C#

Frameworks & Libraries: React, Next.js, Node.js, FastAPI, PyTorch, TensorFlow, OpenCV, NumPy, Pandas, SciPy, Numba JIT, Tailwind CSS, Framer Motion, GSAP, Zustand, Plotly

Datastores & Developer Tools: PostgreSQL, MySQL, Git, Docker, Linux, GitHub Actions, Vercel, Postman